

How the GAVD S-JEPA study evolved
What changed, what improved, and what still remains untested

GAVD S-JEPA tutorial

August 2026

The goal is feature learning, not diagnosis

- ▶ A 64-frame skeleton becomes **528 joint-time tokens**.
- ▶ The student sees visible tokens.
- ▶ A slow teacher sees the complete sequence.
- ▶ A predictor estimates teacher features at hidden locations.
- ▶ The research question is whether the learned vector retains useful gait structure.

Scope: exploratory representation research. Folder labels are not diagnoses verified by this project.

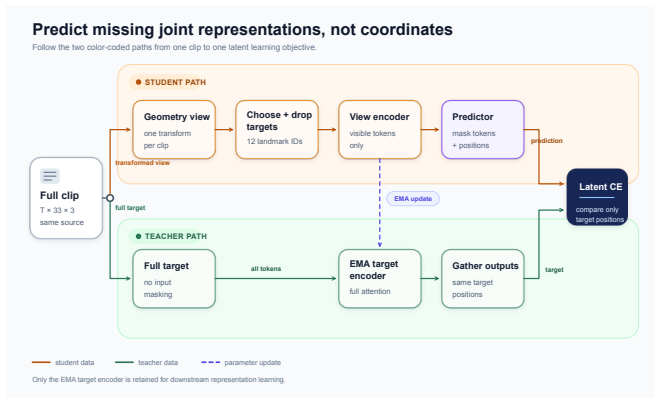
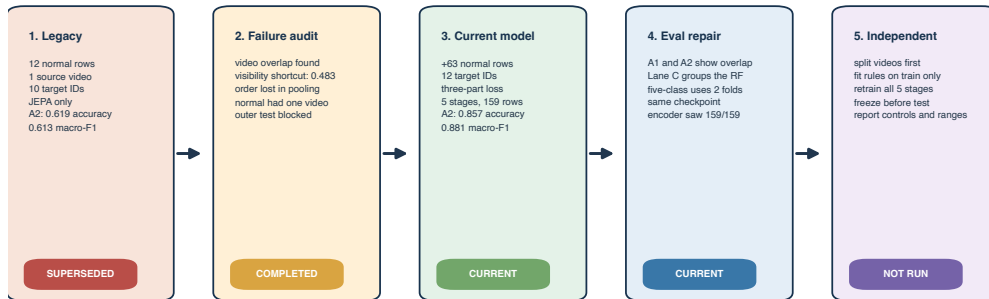


Figure 1: The student predicts hidden teacher features rather than pixels or raw coordinates. Sources: Abdelfattah and Alahi, ECCV 2024; Assran et al., CVPR 2023.

The project improved in three different ways

The project evolved in layers, not through one isolated model change

Completed changes are green or blue. The final card is required work, not a reported result.



Key lesson: a larger in-corpus score, a repaired fold design, and a truly independent estimate are three different kinds of progress.

Figure 2: The evolution separates model changes, evaluation repairs, and work that has not yet been run. Source: current notebooks, artifacts, and result ledger.

The first prototype exposed four limits

Audit finding	Concrete evidence	Response and remaining limit
One normal source	12 windows from 1 video	Add 63 accepted windows from 17 videos, while keeping their provenance separate
Rows were split, not videos	All 16 A1 test videos also occur in classifier training	Add grouped Random Forest audits; the encoder still saw every row
Detector missingness carried labels	Visibility-only A1 accuracy was 0.483	Keep a missingness control beside the learned readout
Mean and standard deviation remove order	The readout compresses all time patches into 384 values	Document the limit; temporal pooling remains a proposed ablation

More normal videos increased breadth and added a provenance risk

Training breadth increased, while provenance became a new confounder

Bar lengths encode sequence counts. Video counts are written beside each stage.

Legacy normal-only



Current Stage 0 normal



Current Stage 4 total



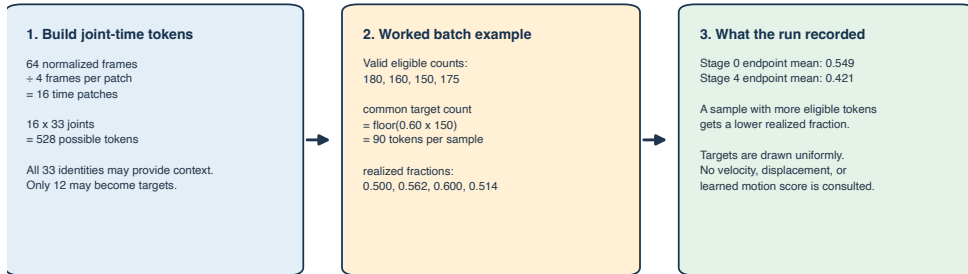
Provenance warning: 63 of 75 normal rows use added extraction; all 84 abnormal rows use canonical extraction.
A classifier may learn camera, crop, bounding-box, or detector differences.
Added labels were not independently clinically reviewed.

Figure 3: Stage 0 grew from one normal video to 18, while the final curriculum reached 159 sequences from 35 videos.

A fixed target rule turns joints into a learnable question

The current masking rule is constrained, uniform, and batch-safe

The numbers in the middle card are a teaching example. Saved run values appear in the final card.



This corrects the misleading shortcut phrase '60% of each sample.' The setting is applied once to the smallest eligible count in the batch.

Figure 4: The configured 0.60 fraction is applied to the smallest valid eligible-token count in the batch.

The objective changed from one job to three

The objective evolved from prediction alone to three explicit jobs



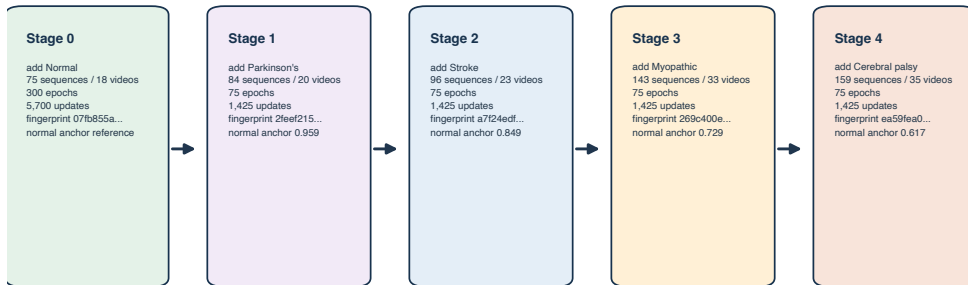
The group term reads condition labels. Only Stage 0 is label-free; Stages 1 through 4 are label-informed representation fine-tuning.

Figure 5: JEPA predicts hidden features, VICReg resists collapse, and the group term shapes condition geometry after Stage 0.

One model continues through five ordered stages

Five fingerprints bind one continuing model lineage

Model and teacher states continue. A fresh optimizer, scheduler, warmup, and EMA schedule position start at each stage.



Total: 600 curriculum epochs and 11,400 optimizer updates. The Stage 4 state is also saved under the final checkpoint alias.

The experiment fingerprint identifies data and configuration lineage. It is not a byte checksum of the checkpoint file.

Figure 6: Model-side state continues, while each stage starts a fresh optimizer and schedule.

Training avoided total collapse, but normal features drifted

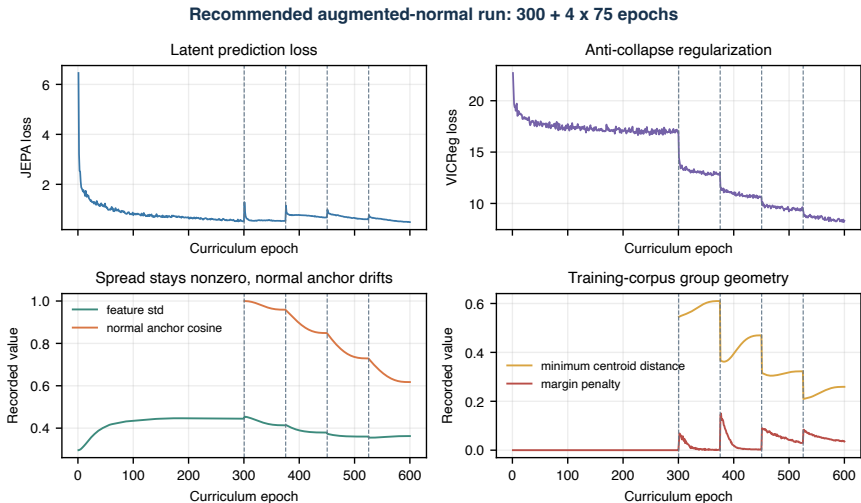


Figure 7: The five-stage run stayed finite, retained nonzero spread, and moved away from its Stage 0 normal reference.

The 384-value summary contains signal, but groups overlap

Centroid geometry is weak, not clean separation

Cosine distance between condition centroids

	Normal	PD	Stroke	Myo	CP
Normal	0.000	0.821	0.334	0.081	0.044
PD	0.821	0.000	0.241	0.546	0.666
Stroke	0.334	0.241	0.000	0.149	0.224
Myo	0.081	0.546	0.149	0.000	0.026
CP	0.044	0.666	0.224	0.026	0.000

0.026

smallest centroid distance

0.104

mean within-group distance

0.054

cosine silhouette

The final encoder saw these rows and their folder labels. This is descriptive geometry.

Figure 8: The smallest condition-centroid distance is lower than the average spread inside a condition.

Three readout lanes answer three limited questions

Three questions need three different claims

A1: all 96

67 / 29 stratified
16 shared videos
29 test rows seen by encoder

Descriptive

A2: exp5 match

47 / 21 legacy split
9 shared videos
21 test rows seen by encoder

Split match only

Lane C: grouped RF

159 rows, 35 videos
binary: 5; five class: 2 folds
all rows seen by encoder

Useful but not independent

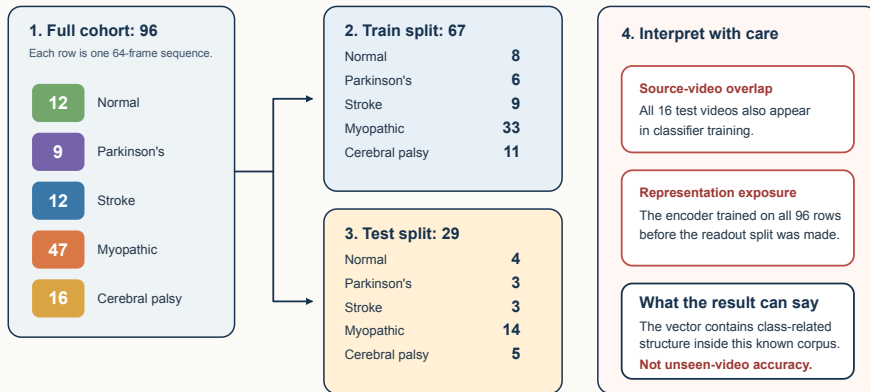
Independent performance still requires fold-local preprocessing choices
and fold-local retraining of all five representation stages.

Figure 9: A1, A2, and Lane C differ in row selection and classifier grouping, but all reuse an exposed encoder.

“All-96 stratified” balances classes, not videos

“All-96 stratified” preserves class proportions in a sequence split

It is useful for a descriptive readout. It is not an unseen-video test.

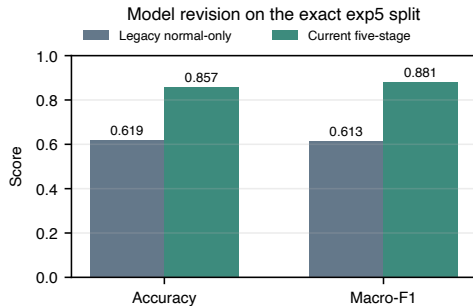


Stratification preserves class proportions. It does not create independent people, videos, cameras, or representation training.

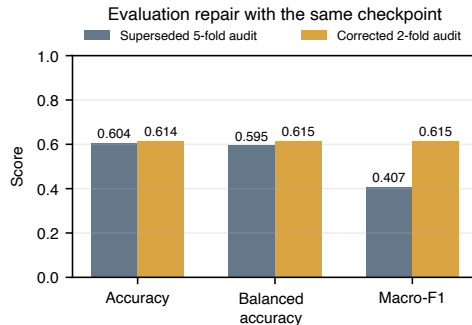
Figure 10: A step-by-step view of the 67-row training split, 29-row test split, and two forms of exposure.

Previous and current scores changed for different reasons

Previous and current results must be compared at the right layer



The model, data, and objective changed; the split stayed confounded.



The model stayed fixed; only the downstream evaluation changed.

Figure 11: The exact-split change used a new model; the Lane C change kept the same model and repaired the fold definition.

Lane C repaired the evaluation, not the model

Lane C improved because its evaluation definition was repaired

The S-JEPA checkpoint and all 159 saved embeddings stayed fixed.

Superseded: 5 ordinary group folds

accuracy 0.604
balanced accuracy 0.595
macro-F1 0.407

One training fold had no cerebral-palsy rows.
Macro-F1 label sets varied by fold.

same encoder
new folds

Corrected: 2 stratified group folds

accuracy 0.614 (change +0.009)
balanced accuracy 0.615 (change +0.020)
macro-F1 0.615 (change +0.207)

Every train and test fold contains all five labels.
Macro-F1 always uses one fixed label order.

Why only two folds? Parkinson's and cerebral palsy have two source videos each.
Two is the largest fold count that keeps every class on both sides.

What did not improve: the encoder was trained once on all 159 rows.
Classifier-level video grouping does not create an independent encoder test.

Figure 12: Two is the largest fold count that keeps every class in both training and testing.

Three symmetry experiments, three different verdicts

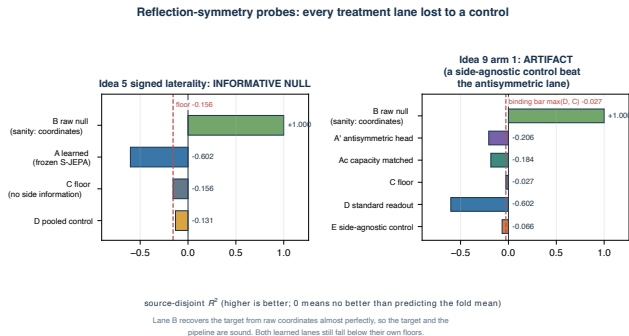


Figure 13: Both readout probes lost to their own controls, and the side-blind lane E outsourced the antisymmetric head.

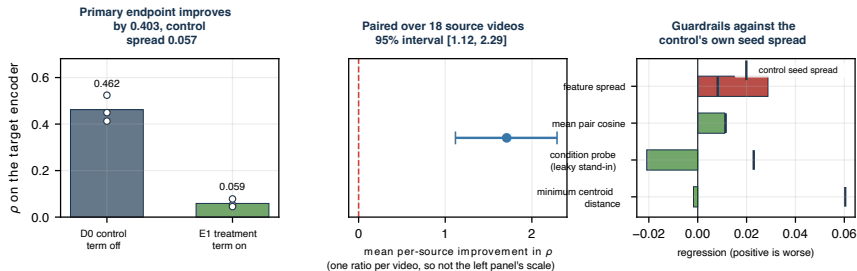
- ▶ **Idea 5, informative null.** The measurement was valid and the answer was no.
- ▶ **Idea 9 arm 1, artifact.** A side-blind control outsourced the treatment, so the claim is withdrawn rather than answered.
- ▶ **Idea 9 arm 2, no credit.** Real on 18 of 18 source videos, but a preregistered guardrail supplies a competing explanation.

Each experiment closed the previous one's escape route.

Two different blockers. On the **labelled** task it is the cohort, not the model or the readout: 18 source videos, and 7.5 percent between-source variance against a preregistered 30 percent, measured in arm 1. For arm 2 it is **lost variance**, which the cohort does not explain.

A reflection-equivariance term reaches the encoder and still earns no credit

Real equivariance ladder, preregistered verdict: **NO CREDIT**



2 of 3 credit conditions met, and all three are required. Failed: condition 3 no guardrail regression.
The head's output scale rose from 0.75 to 1.06, so the endpoint did not improve by the readout collapsing.

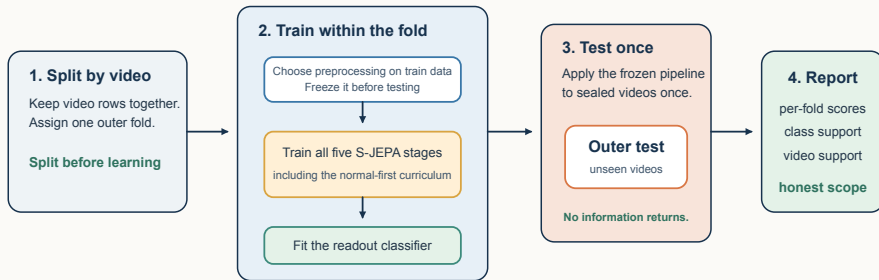
Figure 14: Two of the three preregistered credit conditions passed, and the feature-spread guardrail regressed.

- ▶ **The effect is real:** ρ , where 0 is mirror equivariant and 4 is mirror blind, falls from 0.462 to 0.059, and 18 of 18 source videos improve.
- ▶ **The verdict is still no credit:** feature spread fell from 0.400 to 0.371 against a 0.008 seed spread, so variance loss competes with mirror structure as the explanation.

The next valid test must split videos before learning

The next valid test must retrain the whole pipeline inside each fold

The outer split protects source videos before any learned representation sees them.



Why the current Lane C is only a stress test

Its Random Forest folds separate videos, but the S-JEPA encoder was trained once on all 159 sequences.
A fold-local encoder is the missing step between "grouped classifier" and "unseen-video generalization."

Figure 15: A complete outer fold keeps source videos sealed while preprocessing, all five S-JEPA stages, and the classifier are fitted.

Appendix: legacy and current methods differ at six layers

Legacy and current methods differ at six layers

Layer	Legacy prototype	Current completed run	Impact and remaining limit
Data	12 normal sequences 1 source video	Stage 0: 75 sequences / 18 videos Stage 4: 159 sequences / 35 videos	Broader support, but added-normal provenance can still be a shortcut
Targets	10 eligible landmark identities	12 identities from shoulders through feet uniform batch-safe draw; no motion score	Heels restored; rule is explicit realized fraction is not fixed at 0.60
Objective	Centered, sharpened latent cross-entropy	JEPA + 0.05 VICReg + 0.25 group group term starts after Stage 0	Active anti-collapse pressure added later stages are label-informed
Training	Normal-only checkpoint	Normal first, then four condition stages with condition-balanced replay	One continuing 600-epoch model normal anchor still drifted to 0.617
Artifacts	Loose filename-based reuse	Lineage, cohort, configuration, and fingerprint are recorded	Reject incomplete, stale, wrong-mode, or mixed-fingerprint artifacts
Evaluation	Single video-confounded 47/21 comparison	A1, A2, grouped-RF Lane C, plus missingness and exposure audits	Claims are narrower; no lane yet retrains the encoder outside test videos

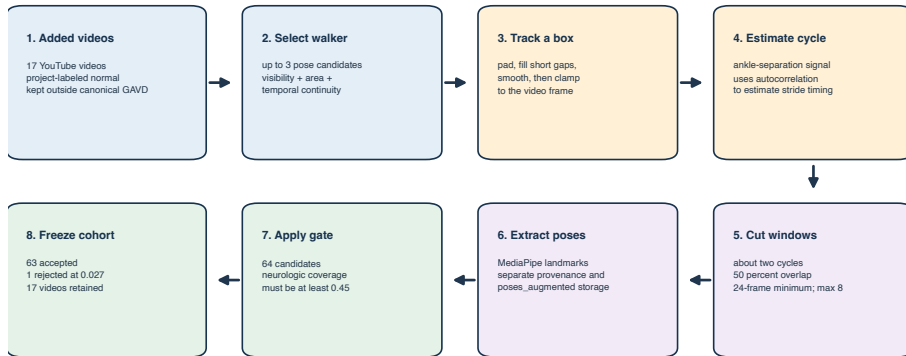
The rows must be interpreted together. The accuracy change cannot be assigned to any single row without a controlled ablation.

Figure 16: Data, targets, objective, training, artifacts, and evaluation all changed together.

Appendix: added normal data use a separate extraction path

Added-normal diversity came from a separate, auditable extraction path

The windows overlap within a video. The scientific gain is 17 additional source videos, not 63 independent people.

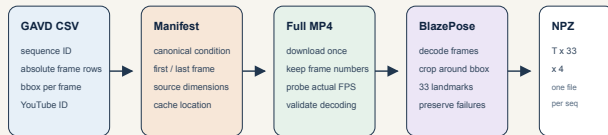


Selection improves reproducibility, but it cannot remove the acquisition-path difference between added normal and canonical abnormal rows.

Figure 17: The added-normal pipeline keeps its candidates, selection rule, coverage threshold, and rejection report separate from canonical GAVD.

Appendix: the full video must remain available

One CSV sequence stays one scientific sample



Why full videos?

Downloading trimmed clips can change frame numbering, breaking the alignment between rows and pixels. The CSV rows refer to the original source timeline, so each video is cached once and decoded in place.

1. Cache the complete source video.
2. Select the annotated frame range.
3. Crop the intended walker.
4. Keep failed pose frames explicit.
5. Center, scale, and resize to 64 frames.
6. Preserve source-video identity for audits.

MediaPipe depth is relative monocular depth, not calibrated 3D motion.

Figure 18: The CSV frame numbers refer to positions in the original source video, so the complete source is cached before clipping and pose extraction. Sources: GAVD and MediaPipe.

Appendix: exact cohort and target identities

Canonical folder label	Sequences	Videos
Normal	12	1
Parkinson's	9	2
Stroke	12	3
Myopathic	47	10
Cerebral palsy	16	2
Total	96	18

Authorized hidden target identities

- ▶ 11, 12: shoulders
- ▶ 23, 24: hips
- ▶ 25, 26: knees
- ▶ 27, 28: ankles
- ▶ 29, 30: heels
- ▶ 31, 32: foot indices

No motion score. No fallback joint. All 33 identities may provide visible context.

Appendix: the 384-value readout is deliberately simple

The current 384-D readout is simple, but it compresses away temporal order

The right side is a proposed controlled ablation. It has not produced a saved result in this repository.

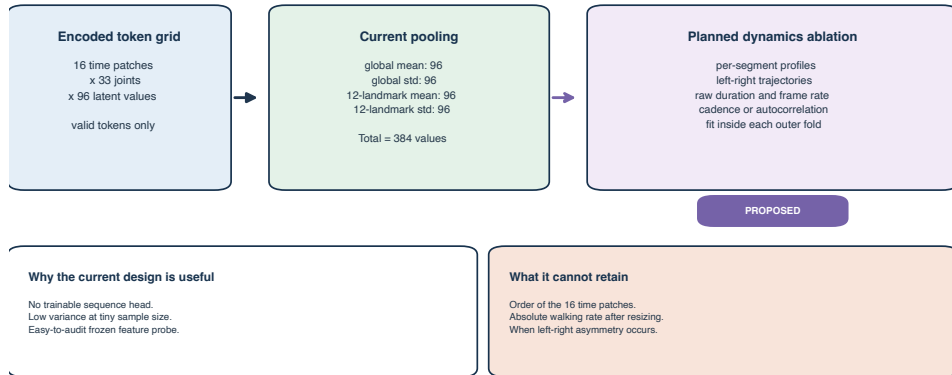


Figure 19: Four 96-value summaries create the current frozen readout; dynamics-preserving alternatives remain proposed.

Appendix: aggregate scores hide class behavior

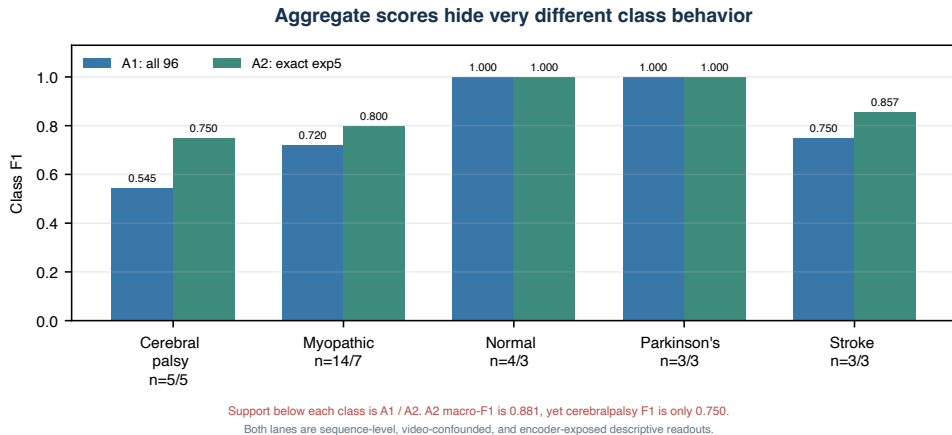
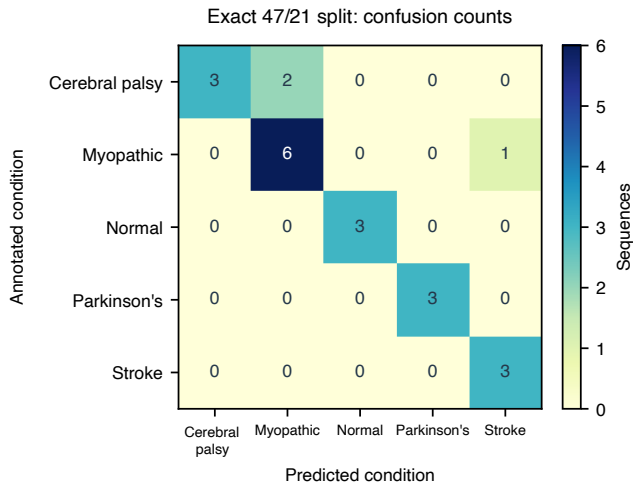


Figure 20: A2 macro-F1 is 0.881, but its cerebral-palsy F1 is 0.750. Support below each class is A1 followed by A2.

Appendix: exact-split errors repeat by source video



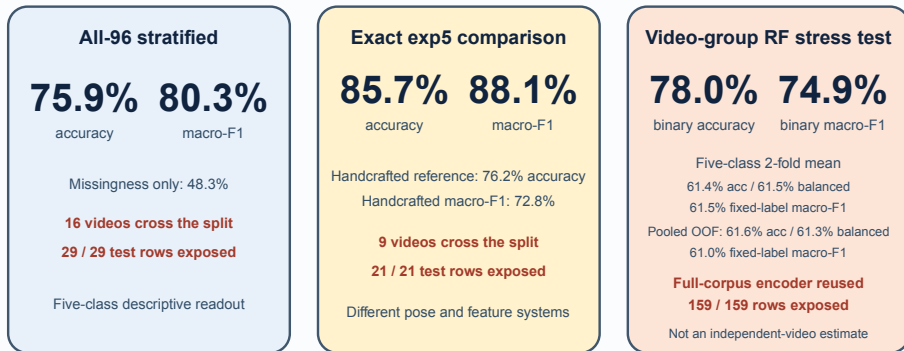
All three errors

- ▶ Two cerebral-palsy sequences from D1PDuHBAP7A were predicted as myopathic.
- ▶ One myopathic sequence from 05o yB0E_0UE was predicted as stroke.

Two of the three errors are repeated windows of one video, so they are not independent mistakes.

Appendix: the current readouts are stress tests

The scores are useful readouts, not held-out clinical performance



The next experiment must retrain the encoder inside each outer video fold.

Figure 22: Every current score has a clear use and a clear exposure warning.

Appendix: model history and current result ledger

Readout version	Accuracy	Balanced accuracy	Macro-F1	Status and meaning
Legacy A2 normal-only	0.619	0.596	0.613	Superseded model; same confounded 47/21 assignment
Current A2 five-stage	0.857	0.891	0.881	Current model; same assignment and encoder exposure
Legacy A1 normal-only	0.621	0.624	0.594	Superseded model; sequence split
Current A1 five-stage	0.759	0.849	0.803	Current model; 29 of 29 test rows exposed
Lane C five-class, old five-fold mean	0.604	0.595	0.407	Superseded evaluation; label support changed by fold
Lane C five-class, corrected two-fold mean	0.614	0.615	0.615	Same checkpoint; every fold has all five labels
Lane C five-class, corrected pooled predictions	0.616	0.613	0.610	Current summary; all 159 rows exposed

Appendix: small binary probes are regression checks

Condition versus normal	Legacy accuracy	Current accuracy	Legacy macro-F1	Current macro-F1	Test rows
Parkinson's	0.714	1.000	0.708	1.000	7
Stroke	0.857	1.000	0.857	1.000	7
Myopathic	0.778	1.000	0.679	1.000	18
Cerebral palsy	0.889	1.000	0.883	1.000	9

Appendix: controls, baselines, and binary Lane C

Check	Accuracy	Balanced accuracy	Macro-F1	Additional detail
A1 missingness only	0.483	0.507	0.477	97 visibility fractions, no coordinates
A2 missingness only	0.286	0.270	0.277	Same control on the 47/21 assignment
A1 majority baseline	0.483	0.200	0.130	Canonical majority is myopathic, 47 of 96
Lane C five-class majority	0.472	0.200	0.128	Lane C majority is normal, 75 of 159
Lane C normal versus abnormal	0.780	0.804	0.749	Five grouped RF folds; mean ROC AUC 0.915

Appendix: every preregistered symmetry gate, resolved

Preregistered gates, resolved

2 of 10 gates passed, and every one of those is a wiring or control check rather than a result. That is what makes the negative verdicts trustworthy rather than a bug.

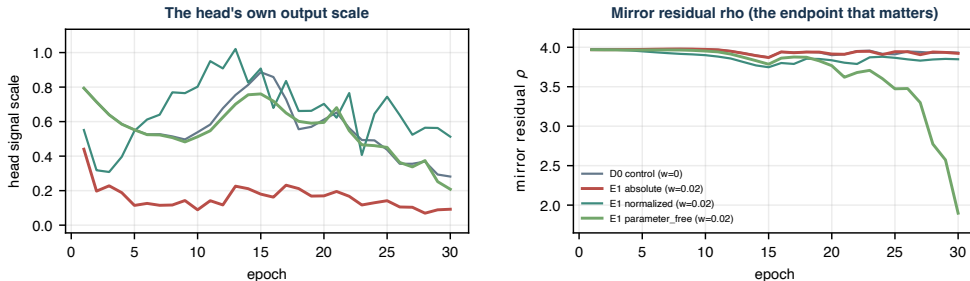
Idea 5	beats floor by 0.05	not met
Idea 5	reaches 80% of raw null	not met
Idea 5	correct sign on 75% of held-out sources	not met
Idea 5	pooled control behaves	PASSED
Idea 9	beats binding bar by 0.05	not met
Idea 9	beats capacity-matched by 0.05	not met
Idea 9	better than permuted labels	not met
Idea 9	side-agnostic lane stays weak	not met
Idea 9	target varies between sources	not met
Idea 9	head wiring verified	PASSED

Only 7.5% of the target's variance lies between source videos against a 30% requirement, so source-disjoint folds have very little shared signal to find.

Figure 23: Two of ten preregistered gates passed, and both are wiring or control checks rather than results.

Appendix: a loss term that satisfied itself instead of the encoder

Why the original equivariance term had to be replaced



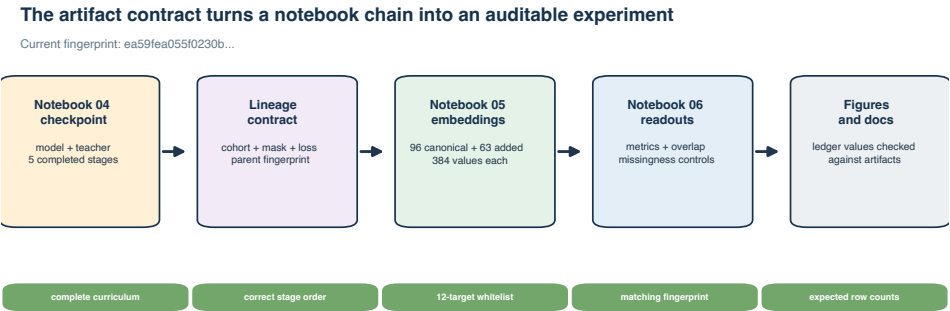
Synthetic fixtures, not gait. Rho is 0 when mirror equivariant and 4 when mirror blind. The red absolute form cuts its own loss

184-fold by shrinking the head 4.8-fold (left), yet leaves the endpoint within 0.010 of the control (right).

Both scale-invariant forms clear the gate, but parameter_free moves rho by 2.03, so it is the one carried forward.

Figure 24: Synthetic fixtures, not gait: the absolute form shrinks the head's output while the mirror residual stays flat, and a scale-invariant form removes that escape route.

Appendix: the artifact contract rejects silent substitution



A missing filename is no longer silently replaced by a different checkpoint. Consumers stop when the lineage or cohort does not match.

Figure 25: Checkpoint lineage, embeddings, readouts, and figures must agree on the completed curriculum and experiment fingerprint.

Appendix: seven notebooks form one audit trail

Build understanding and evidence in seven steps

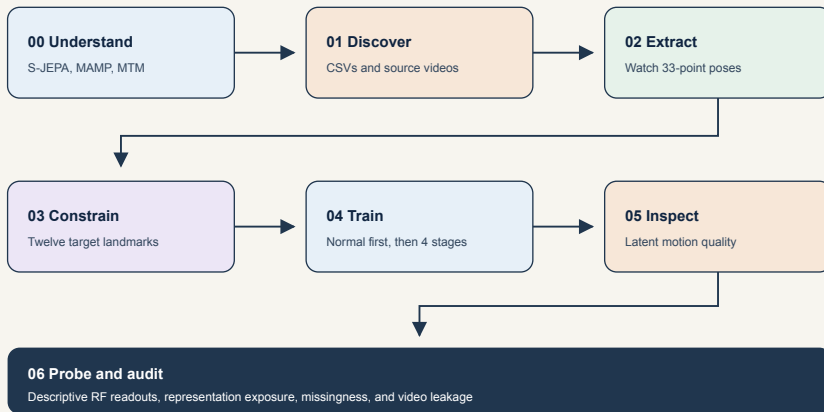


Figure 26: Each notebook produces an artifact that the next notebook can verify.

Appendix: reproduce and verify

Primary artifacts

- ▶ `sjepa_curriculum_final_augmented.pt`
- ▶ `curriculum_training_history_augmented.csv`
- ▶ `curriculum_stage_summary_augmented.csv`
- ▶ `sequence_embeddings.parquet`
- ▶ `classifier_metrics.csv`
- ▶ `lane_c_video_disjoint_metrics.csv`
- ▶ `classifier_contract.json`
- ▶ `result_history.csv`

Current experiment identity

`ea59fea055f0230bcf236deb1d1e8bbf08033766`

`e7cd95a98f28210b3042c4e4`

The fingerprint names data and configuration lineage. It is not a byte checksum.

Run notebooks 00 through 06 in order. Use real mode for reported values.

Appendix: separate completed work from proposed work

What is implemented, proposed, or explicitly rejected

Implemented and measured	Proposed next experiments	Rejected interpretations
• 12-landmark fixed target whitelist • uniform batch-safe masking with validity gates • 63 accepted added-normal pose sequences • five stages with replay, VICReg, and group loss • fingerprinted checkpoints, embeddings, and result ledger • A1, A2, and encoder-transductive Lane C audits	• outer source-video split before all fitting • fold-local five-stage representation training • source-balanced acquisition and provenance controls • dynamics and raw-rate readout ablations • same-pose handcrafted, embedding, and fused comparison • uncertainty from genuinely independent outer folds	• Lane C is an unseen-video encoder estimate • every sample masks exactly 60 percent • added normal labels are canonical GAVD labels • VICReg guarantees clean clinical clusters • binary and five-class bar heights are comparable • a fixed Random Forest can undo encoder exposure

The archived improvement plan remains useful for hypotheses, but only the green column describes the completed checkpoint.

Figure 27: Only the green column describes the completed checkpoint.

Appendix: references and responsible use

1. Abdelfattah and Alahi. S-JEPA. *ECCV*, 2024. [DOI](#)
2. Assran et al. I-JEPA. *CVPR*, 2023. [DOI](#)
3. Ranjan et al. GAVD. *IEEE Access*, 2025. [DOI](#)
4. Grishchenko et al. BlazePose GHUM Holistic, 2022. [DOI](#)
5. Bardes, Ponce, and LeCun. VICReg. *ICLR*, 2022. [OpenReview](#)
6. Google AI Edge. Pose Landmark Detection Guide. [Documentation](#)
7. Roberts et al. Group-aware cross-validation. *Ecography*, 2017. [DOI](#)
8. Bengio and Grandvalet. Variance of K-fold cross-validation. *JMLR*, 2004. [Article](#)
9. Rousseeuw. Silhouettes for cluster analysis, 1987. [DOI](#)
10. Breiman. Random Forests. *Machine Learning*, 2001. [DOI](#)
11. Hui et al. Motion-topology masked prediction. *Scientific Reports*, 2026. [DOI](#)

Responsible use: research comparison only. Do not use these outputs for diagnosis, triage, or patient-level decisions.